

TAYLORVILLE, IL, USA

WMO: 744662

Lat: 39.534N

Lon: 89.328W

Elev: 622

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 63817

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.0	8.4	-6.1	4.1	4.8	-0.5	5.5	9.7	28.2	30.6	25.6	28.2	7.5	320	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	93.4	76.0	91.0	75.2	89.2	74.5	80.1	89.3	78.3	87.4	76.8	85.4	7.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	146.1	85.8	75.4	136.1	83.0	73.9	129.5	80.9	44.0	88.0	42.4	88.0	40.5	86.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.4	20.8	18.8	-5.4	97.0	6.1	5.7	-9.8	101.0	-13.4	104.3	-16.8	107.5	-21.3	111.6
			-6.1	82.7	6.0	1.4	-10.4	83.7	-13.9	84.5	-17.3	85.2	-21.7	86.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	27.9	30.5	43.6	54.6	65.1	73.3	75.7	73.9	68.7	55.9	43.1	33.1
	DBStd	19.33	11.84	11.43	11.43	9.66	8.69	5.78	5.88	5.92	7.54	9.63	10.01	9.81
	HDD50	2400	690	552	267	61	4	0	0	0	46	252	528	
	HDD65	5252	1151	966	667	332	112	7	3	5	47	315	658	989
	CDD50	3820	3	6	69	198	471	700	795	742	560	228	44	4
	CDD65	1199	0	0	4	19	114	257	333	282	157	32	1	0
	CDH74	11463	0	0	34	207	991	2419	3331	2619	1553	303	6	0
	CDH80	4429	0	0	2	46	298	929	1409	1053	608	84	0	0
Wind	WSAvg	8.3	10.4	9.9	10.4	10.8	8.7	6.7	5.0	4.8	5.6	7.9	9.5	9.8
Precipitation	PrecAvg	37.8	2.4	1.7	2.6	3.8	4.7	4.4	3.3	3.3	2.8	3.4	3.2	2.2
	PrecMax	48.3	6.6	5.1	5.0	9.3	10.7	8.4	7.7	5.3	9.1	6.6	7.6	6.5
	PrecMin	30.1	0.5	0.5	0.9	1.6	1.3	2.2	1.4	0.8	0.1	0.9	0.4	0.7
	PrecStd	5.2	1.4	1.1	1.1	1.9	2.7	1.8	1.7	1.0	2.0	1.6	1.7	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.1	64.3	79.1	84.5	90.9	93.5	100.1	99.0	96.6	87.8	73.5	62.7
		MCWB	57.0	56.5	64.0	63.7	72.7	75.6	77.8	77.9	72.9	70.5	61.4	58.8
	2%	DB	54.8	59.0	72.3	80.7	86.4	91.0	93.3	93.1	90.4	81.8	67.5	56.5
		MCWB	51.7	54.2	61.3	62.4	70.4	74.7	77.8	75.7	72.1	66.5	59.4	52.8
	5%	DB	49.6	53.1	66.0	74.9	82.7	89.1	90.7	90.1	87.7	76.8	63.1	52.3
		MCWB	46.6	48.6	58.1	60.9	68.1	74.0	77.2	75.5	71.7	64.1	56.3	49.3
	10%	DB	43.5	46.4	61.5	71.7	80.6	85.7	88.0	86.0	82.4	72.1	58.8	46.3
		MCWB	40.0	42.5	54.9	59.2	66.8	72.4	76.0	74.0	68.5	62.1	53.0	43.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.2	59.2	66.1	68.5	75.7	80.3	82.6	81.9	77.6	72.9	65.6	59.0
		MCDB	60.9	63.2	75.4	79.3	86.7	88.9	93.1	92.1	88.8	82.7	71.4	62.1
	2%	WB	52.7	54.9	62.6	65.7	73.1	78.0	80.6	79.6	75.5	70.2	61.3	54.3
		MCDB	54.7	57.7	69.5	73.0	82.8	87.3	90.1	88.8	84.5	77.8	65.7	56.0
	5%	WB	46.5	48.3	59.6	63.3	71.0	76.1	79.0	77.6	73.7	67.2	57.7	48.9
		MCDB	48.4	52.4	65.3	71.1	79.5	85.3	87.7	86.8	81.7	72.8	62.1	51.6
	10%	WB	40.4	42.9	55.7	60.7	69.0	74.4	77.4	75.7	71.7	64.0	53.5	44.6
		MCDB	44.0	46.8	60.9	69.1	76.7	82.7	85.8	83.2	79.5	70.3	57.8	46.5
Mean Daily Temperature Range	5% DB	MDBR	14.9	15.9	18.1	21.0	20.1	20.5	20.3	20.9	23.7	21.4	17.3	13.5
		MCDBR	21.9	23.2	24.4	26.3	23.8	23.7	23.0	24.7	26.2	26.0	23.2	19.4
		MCWBR	18.5	18.4	15.3	13.6	10.5	10.0	9.4	10.2	9.9	12.4	16.0	16.3
	5% WB	MCDBR	20.7	22.1	21.3	22.4	20.1	20.3	20.0	20.3	21.3	21.4	20.8	18.2
		MCWBR	18.6	18.6	14.2	14.5	10.3	9.9	9.9	9.9	9.8	12.1	16.7	16.7
Clear-Sky Solar Irradiance	taub		0.302	0.313	0.349	0.391	0.431	0.429	0.460	0.449	0.407	0.357	0.310	0.297
	taud		2.453	2.411	2.386	2.269	2.214	2.254	2.156	2.176	2.275	2.421	2.505	2.483
	Ebn at Noon		276	288	287	280	270	270	260	260	264	268	271	269
	Edh at Noon		25	30	35	42	45	43	47	45	38	30	24	23
All-Sky Solar Radiation	RadAvg		597	861	1163	1541	1744	2005	1981	1761	1488	1039	695	502
	RadStd		70	80	88	94	144	148	129	114	102	115	69	63

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	+140

Nomenclature: See separate page